Project Assignment 1

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CSE5825 - BML

Dataset Description

Motivation

Today, 97% of adults say they use the internet [1], with social media and online forums central to how people communicate, share information, and express opinions. These platforms have continued to grow and produce user-generated content that offers insights into online communities and overall public opinion.

However, the authenticity of online content is shifting as AI-generated content and automated accounts (bots) overtake human activity. Recent analysis shows that signs of AI authorship are present in over one-third of pages published since the release of ChatGPT [2]. Furthermore, machine-generated web traffic has officially surpassed human traffic, with bots now accounting for 64.2% of all HTTP requests to HTML content [3]. The increasing presence of AI-generated content and bot traffic raises questions about the authenticity of discourse and engagement online, and also about how public sentiment towards AI has changed.

Dataset Overview

Reddit is a popular platform for online discussion, used by 26% of adults in the US [4]. Self-described as "the most real place on the internet," users can create posts, write comments, and vote on submissions using upvotes and downvotes. The content on Reddit ranges from news articles to niche interests to anonymous venting posts, and is organized into smaller subforums called subreddits. Each subreddit is a microcosm with its own rules, moderation, and community culture.

The dataset for this project is the open-index/arctic dataset on Huggingface [5], consisting of all posts and comments on Reddit since 2005 from every publicly available subreddit. The HuggingFace dataset reformats the data from Project Arctic Shift, which is an open-source data archiving project that saves web scrapings from Reddit as monthly Zstandard dumps [6]. The dataset is 1.3 TB and has 15.7 billion total entries, with 12.9 billion comments and 2.8 billion posts. Posts and comments are stored as separate datasets and split into Parquet shards.

Data fields for both posts and comments include the Reddit-generated ID, the author, subreddit to which it was posted, content body, content title for posts, timestamp of creation, and score (number of upvotes – can be positive or negative). Comments are linked to parent comments and submissions via a parent ID and link ID, respectively.

Dataset Curation

Two subsets were created from the open-index/arctic dataset for data visualization and analysis. The first subset contains Reddit submissions from 2010-2026 across 5 popular subreddits in each of seven generalized categories:

News and Politics	Discussion and Support
Technology and Science	Media and Entertainment
Business and Finance	Lifestyle, Hobbies and Living
Education and Career	

The starting year of 2010 was selected because subreddit data was not available before 2009. For each year, 400 submissions were randomly sampled from each of the 5 subreddits per category. Half of the submissions were randomly sampled from posts with a score of at least 50 to ensure some degree of engagement, while the remaining half were randomly sampled without a requirement.

The second subset consists of all submissions from 5 popular subreddits from the News and Politics category, specifically r/politicaldiscussion, r/politics, r/worldnews, r/Conservative, and r/conspiracy between April 2025 and February 2026, the most recent period available in the dataset. This subset provides a more granular view of activity and engagement within these subreddits over a more recent period.

Biases and Limitations

The dataset has several potential biases and limitations that should be considered when interpreting the results. Reddit users do not represent the general population, and its usage varies across demographic groups such as race, gender, and age [4]. Therefore, the opinions and activity expressed on Reddit might not reflect the broader public. Beyond these platform-level biases, the dataset curation process also introduces potential sampling bias. Since the subreddits were selected based on current popularity, communities that were popular in earlier years but have since declined in activity might be underrepresented. This introduces survivorship bias because the selected subreddits might not reflect the communities that were the most active throughout the entire 2010-2026 period. Additionally, selecting only 5 subreddits per category excludes smaller niche communities that might have different perspectives or activity.

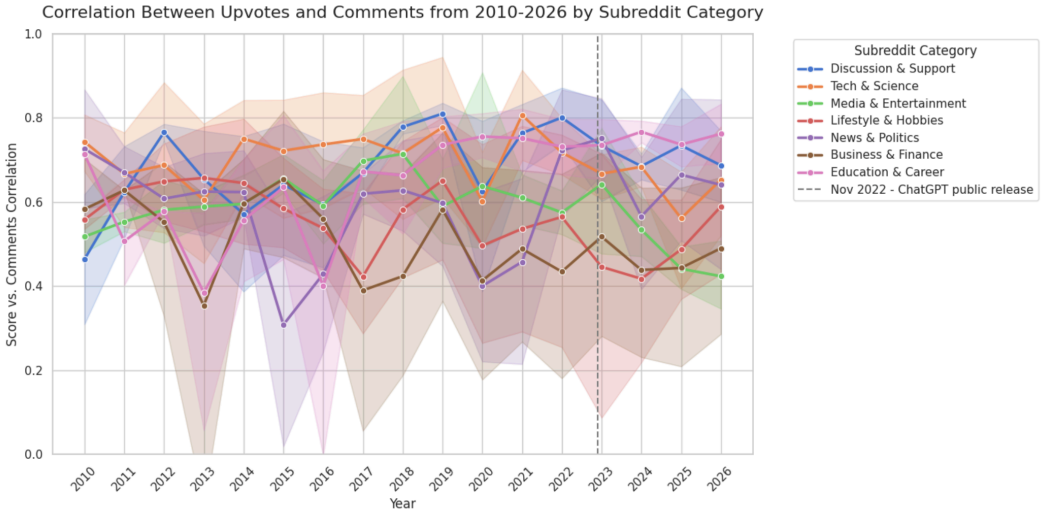
The dataset's structure and available fields also introduce limitations to the analysis. The dataset does not include subreddit information for entries before 2010, limiting the time range available for any temporal subreddit analysis. The original dataset also lacks comment data after August 2022. This limits the ability to analyze changes in discussion and engagement following the release of ChatGPT in November 2022. Finally, the dataset also lacks detailed account information and user history, making it difficult to distinguish between authentic users and automated accounts based on account behavior alone.

Data Visualization

1. Correlation Between Upvotes and Comments from 2010-2026 by Subreddit Category

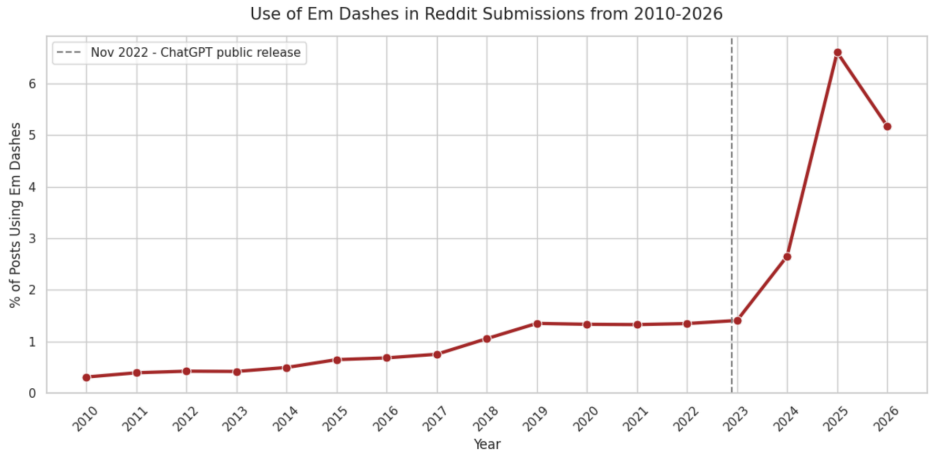
This visualization shows how the correlation between upvotes (post scores) and comment counts has changed over time across subreddit categories from 2010-2026. A stronger positive correlation indicates that posts with more upvotes generally also receive more comments. Some interesting observations:

- Noticeable dips in correlation in News & Politics subreddits during election years (2015/2016, 2020, 2024), potentially indicating anomalous engagement patterns.
- Consistently strong correlation in the Discussion & Support category across the years, indicative of active conversation and engagement.
- Relatively low correlation in Business & Finance compared to other categories throughout the years.



2. Use of Em Dashes in Reddit Submissions from 2010-2026

This visualization displays the percentage of Reddit text posts containing em dashes (–) from 2010 to 2026 across various subreddits. Em dashes are a common feature of LLM-generated text. Per the visualization, there is a noticeable uptick in the use of em dashes in Reddit posts after ChatGPT's release. More generally this shows a change in the writing characteristics of Reddit submissions over time.

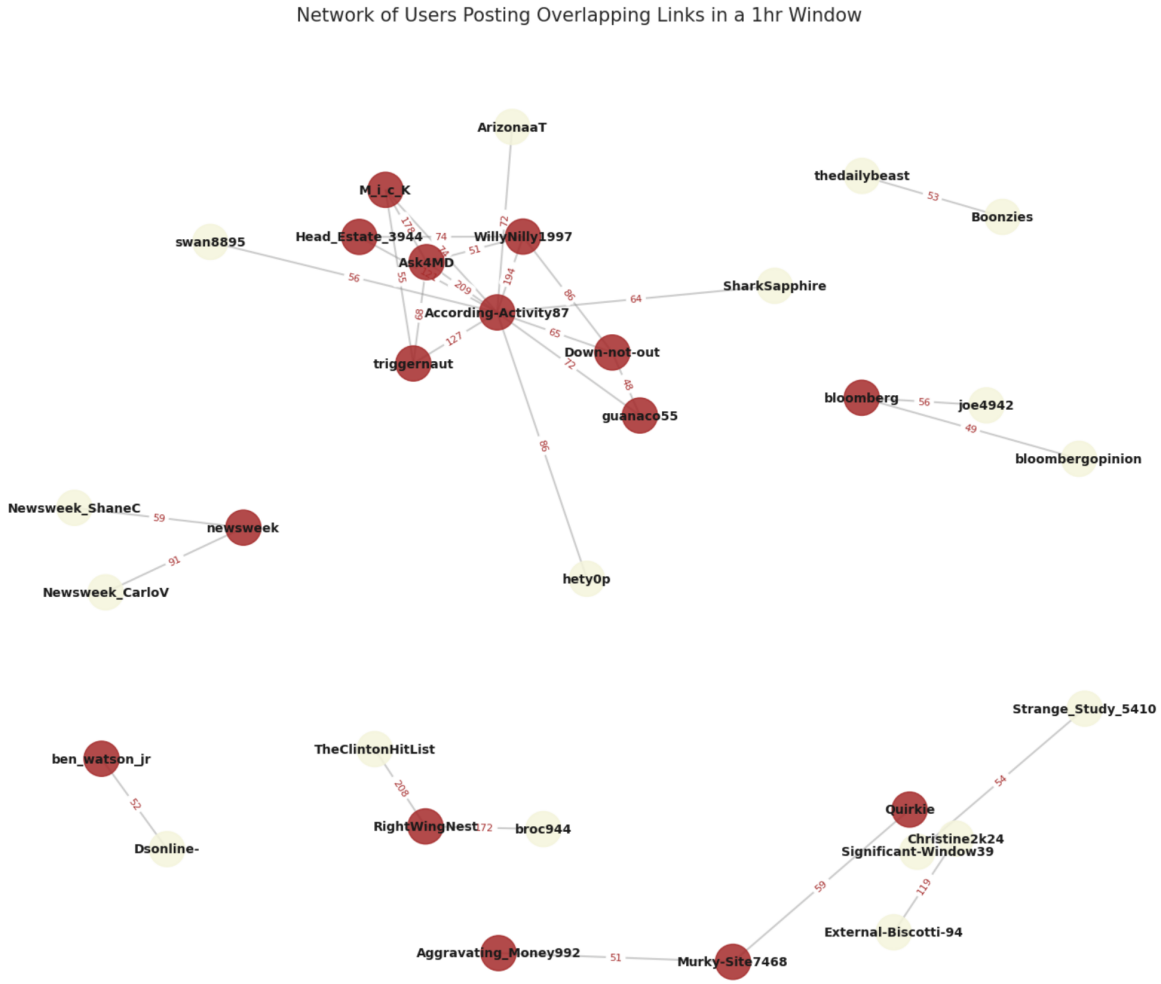


3. Graph Network of Users Posting Overlapping Links in a 1 Hour Window

This visualization shows overlapping post activity between Reddit users across 5 popular political subreddits from April 2025 to February 2026. This subset of data contains 371,178 unique posts from 83,320 users. The 30 most frequent posters accounted for 40,873 posts, or 11.32% of all posts during this period, highlighting how posting activity was concentrated among a relatively small group of users (0.036% of the total users).

Nodes on the graph represent individual users, and edges connect users who submitted links from the same base domain within a one-hour window. Edge weights indicate the number of overlapping submissions. The higher the edge weight, the more post overlap between 2 users. Additionally, nodes are colored brown if the user was found to be among the 30 most frequent posters and beige otherwise.

This graph reveals patterns of overlapping activity between users rather than just post frequency. Of the 30 most frequent posters, 15 of them had overlapping submissions and appeared in the graph. These 15 users accounted for 27,426 total posts, representing 7.59% of all posts across the 5 subreddits.



Modelling the Data

Observed Variables

The observed variables in the data include the raw text contents of the Reddit posts and comments, along with their respective metadata. The timestamp and created date of the content are both temporal variables. Engagement is represented by discrete numeric variables, specifically the score (net upvote and downvotes), and the comment count for submissions. The subreddit, or community where the content was posted, is a categorical variable, along with the content author and a unique ID. Additional categorical variables are also used to link posts and comments: the parent ID variable captures any nesting, and the link ID variable associates each comment with its original post.

Relationships Between Variables

The post score and comment count are expected to be positively correlated, as submissions that receive more upvotes might also generate more discussion. However, this relationship can vary across subreddit categories and over time, as seen in the first visualization.

The timestamp and subreddit variables could also be associated with differences in content characteristics. Specifically, the text content of a post could have a relationship with the timestamp. For example, the second visualization shows an increase in the percentage of submissions containing em dashes following the release of ChatGPT.

The author and timestamp variables can also be used to examine posting frequency and overlapping activity between users. For example, users who frequently submit posts may be more likely to share links from the same domains within a short time window. In the third visualization for example, posting activity is concentrated among a relatively small group of users, and some of them exhibit similar posting patterns.

Latent Variables

The hidden/latent variables represent underlying characteristics of Reddit content and activity that isn't directly captured by the observed variables. The following latent variables can be introduced:

- Manufactured Popularity: this refers to activity that could be artificially generated or manipulated to increase the visibility of certain content. Patterns in posting frequency, post scores vs comments, authors, and similar activity between users could potentially provide insight into coordinated/automated engagement.
- Sentiment: this represents the underlying attitudes or opinions expressed in Reddit content. Given the raw text, which is an observed variable, the post sentiment can be inferred from the language used and the context.
- LLM-Generated Content: this latent variable represents whether Reddit content was generated by AI vs. written by a human. Linguistic patterns and key characteristics of LLM writing can be used as identifiers to determine whether or not the content appears to be AI generated.

Hypotheses

Based on the latent variables defined above, the following three hypotheses may be supported:

1. Identifying Manufactured Popularity/Astroturfing
   Do posts in political, news, technology, and finance subreddits tend to have more abnormal engagement than posts in more casual subreddits such as hobby and general discussion subreddits?
2. Changes in Discourse on AI Over Time
   Do technology subreddits display an increase in negative or skeptical language towards AI over time?
3. Detecting AI-Generated Content Across Subreddits
   Do certain subreddits contain a higher proportion of posts and comments that appear to be written by AI? For example, posts with little sentence variety, repetitive phrasing, and overuse of buzzwords.

References

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